

Target validation paths — pancreatic-recurrent-kras-g12r-m8f3

The diagnostic and biomarker workup that hardens the targetable-feature call

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Target validation paths

All six biomarkers in this case read as confirmed on the local report, so the workup here hardens the calls rather than gates a fresh diagnosis. The single essential row is orthogonal KRAS G12R confirmation by plasma ctDNA plus a second-platform tissue NGS: pan-RAS and RAS(ON) sponsors re-test the variant on plasma at screening regardless of the local report, and the same draw seeds the baseline VAF used for on-treatment monitoring at cycle 2 and cycle 4. This re-test gates trial entry for daraxonrasib (RMC-6236, NCT05379985), the daraxonrasib + chemo combination platform (NCT06445062), and the zoldonrasib (RMC-7977) + ivonescimab program (NCT07397338). Five high-priority workups run in parallel: a comprehensive germline panel, p16 IHC on the CDKN2A call, CCND3 alteration-class refinement, KRAS co-mutation NGS (SMAD4 / KEAP1 / STK11 / GNAS / ARID1A), and MTAP reflex testing on the 9p21 locus. Six medium-priority workups round out the dossier.

KRAS G12R

Essential: orthogonal NGS plus baseline ctDNA. Pan-RAS / RAS(ON) trial central labs re-test KRAS G12X on plasma at screening; the orthogonal call locks eligibility for daraxonrasib monotherapy (NCT05379985), daraxonrasib + chemo (NCT06445062), and the RMC-7977 + ivonescimab combination (NCT07397338). Reference platforms are Guardant360 CDx, FoundationOne Liquid CDx, Tempus xF+, Natera Signatera, and Caris Assure; the FDA-approved CDx platforms are the safest choice for trial-entry documentation. G12R is biologically distinct from G12C and G12D (RAS-GTP-binding-deficient per Hobbs 2020), so G12C-specific covalent agents are not the relevant class. Turnaround is one to two weeks.

Three high-priority companions ride on the same NGS order. Co-mutation profiling for SMAD4, KEAP1, STK11, GNAS, and ARID1A frames prognosis and weights any ICI-combination decision: SMAD4 loss in PDAC is the strongest single co-alteration linked to wider metastatic spread, and KEAP1 / STK11 predict diminished ICI benefit on KRAS-mutant backgrounds. A germline hereditary-cancer panel (BRCA1, BRCA2, PALB2, ATM, MLH1, MSH2, MSH6, PMS2, EPCAM, CDKN2A, TP53, STK11) is NCCN-recommended for every PDAC patient regardless of family history. A pathogenic BRCA1 / BRCA2 or PALB2 result opens olaparib maintenance via NCT02184195 plus rucaparib as the platinum-induction-gated PARP follow-on. Germline turnaround is three to six weeks; order in parallel with a genetic-counseling referral.

Two medium-priority resistance reads sit alongside: KRAS copy-number annotation (baseline KRAS amplification and acquired wildtype-KRAS amplification are documented escape mechanisms on the pan-RAS class), and serial KRAS G12R ddPCR or tumor-informed MRD at cycle 2 and cycle 4 of any KRAS-directed therapy, which flags VAF rebound weeks ahead of RECIST imaging.

TP53

The local report annotates an inactivating mutation, but the functional class matters: pure loss-of-function (truncations, splice) reads differently from dominant-negative missense hotspots (R175H, R248W, R273H) and from gain-of-function variants, and any TP53-restoration or MDM2-axis strategy gates on the precise call. The workup is a same-day re-read of the existing NGS report against the IARC TP53 database for variant annotation in HGVS notation. Does not gate the KRAS axis; refines any downstream cell-cycle / DDR conversation.

CDKN2A

Two confirmatory reads. p16 IHC (clone E6H4 on archival FFPE, one to two week turnaround) is the orthogonal protein-level check on the NGS copy-number call; CDK4/6-inhibitor trial central labs prefer p16-negative status or biallelic CDKN2A loss for enrollment in CDKN2A-loss-enriched arms. A discordance (NGS-loss but IHC-retained) suggests subclonal or assay-driven calls and tempers expectations for CDK4/6 monotherapy.

Multi-region p16 IHC on a second tumor block (primary plus metastatic site if available) reads PDAC clonal heterogeneity at the next tier. A patchy result does not gate enrollment, but it changes how a CDK4/6-inhibitor response is interpreted and may push the board toward combination strategies.

The 9p21 MTAP reflex (callable from the same NGS panel) is the other operational read: CDKN2A and MTAP are co-deleted in roughly 80 to 90 percent of CDKN2A homozygous deletions, and MTAP co-deletion is the eligibility threshold for the MTA-cooperative PRMT5 class (anvumetostat on NCT06360354, BMS-986504 on NCT07492680). Decisive for any decision on whether the dual-feature daraxonrasib + anvumetostat regimen is on the table.

CCND3

The local NGS report names a CCND3 alteration without specifying class. An amplification or activating mutation supports cyclin-D-axis dependency and CDK4/6 sensitivity; a passenger missense without copy gain carries weaker mechanistic weight. Cyclin-D-degrader and selective CDK4 programs (PF-07220060, fadraciclib, INCB123667) gate on copy-number gain or known activating mutations, so the class call frames whether these axes belong on the page at all. Usually returnable as an annotation re-read of the existing NGS panel; ask the molecular pathologist for the alteration class explicitly. Turnaround is two to three weeks if a fresh order is required.

MSS

Two reads to confirm the negative ICI finding. MMR IHC (MLH1, MSH2, MSH6, PMS2 on archival FFPE, one to two week turnaround) is the orthogonal protein-level confirmation on the NGS call. NGS-derived MSI calls have known false negatives at low tumor purity, and most ICI-trial central labs accept MMR IHC on its own. A confirmed MSS by both NGS and IHC closes the door cleanly on the tumor-agnostic pembrolizumab pathway.

The tumor microenvironment workup (PD-L1 22C3, CD3 / CD8 TIL density, alpha-SMA or H and E stromal density) frames any ICI-combination strategy. PDAC is the textbook immune-cold, stromally dense tumor, and the dominant interpretive question is usually T-cell exclusion rather than PD-L1 score. Returnable on a multiplex IHC panel (NeoGenomics MultiOmyx covers PD-L1, CD3, CD8, and stromal markers on a single section).

TMB

Documentation review rather than a new order. TMB values are sensitive to panel size, mutation-call filters, and whether synonymous variants are included; 4.1 mut/Mb on a 500-gene panel may map differently against the FDA-approved tumor-agnostic pembrolizumab threshold than a whole-exome-derived value (Friends of Cancer Research TMB Project harmonization study). If the existing report is from FoundationOne CDx, this is a same-day documentation check on the bioinformatics methods section. Below the 10 mut/Mb threshold the tumor-agnostic pembrolizumab route is foreclosed; the read confirms that foreclosure cleanly.

Where to order these assays

Assay	Provider	Decision gated	Contact
Orthogonal NGS panel plus ctDNA (KRAS G12R-specific ddPCR or comprehensive plasma NGS)	Guardant Health (preferred) (Guardant360 CDx)	Pan-RAS / RAS(ON) inhibitor trials (RMC-6236 NCT05379985, RMC-7977 NCT06445062); G12C-specific agents are not the relevant class.	test info · 505 Penobscot Drive, Redwood City, CA 94063 · 1-855-698-8887
Orthogonal NGS panel plus ctDNA	Foundation Medicine (FoundationOne Liquid CDx)	Pan-RAS / RAS(ON) inhibitor trials (RMC-6236 NCT05379985, RMC-7977 NCT06445062); G12C-specific agents are not the relevant class.	test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639
Orthogonal NGS panel plus ctDNA	Tempus (Tempus xF+ (plasma) / xT (tissue))	Pan-RAS / RAS(ON) inhibitor trials (RMC-6236 NCT05379985, RMC-7977 NCT06445062); G12C-specific agents are not the relevant class.	test info · 600 West Chicago Avenue, Chicago, IL 60654 · 1-800-739-4137
Orthogonal NGS panel plus ctDNA	Natera (Signatera (tumor-informed MRD))	Pan-RAS / RAS(ON) inhibitor trials (RMC-6236 NCT05379985, RMC-7977 NCT06445062); G12C-specific agents are not the relevant class.	test info · 13011 McCallen Pass, Austin, TX 78753 · 1-650-249-9090
Orthogonal NGS panel plus ctDNA	Caris Life Sciences (Caris Assure)	Pan-RAS / RAS(ON) inhibitor trials (RMC-6236 NCT05379985, RMC-7977 NCT06445062); G12C-specific agents are not the relevant class.	test info · 4610 South 44th Place, Phoenix, AZ 85040 · 1-888-979-8669
p16 IHC (clone E6H4 or equivalent)	Mayo Clinic Laboratories (preferred)	CDK4/6-inhibitor eligibility (palbociclib, ribociclib, abemaciclib) and CDKN2A-loss-enriched trial arms.	test info · 200 First Street SW, Rochester, MN 55905 · 1-800-533-1710
p16 IHC (clone E6H4 or equivalent)	ARUP Laboratories	CDK4/6-inhibitor eligibility (palbociclib, ribociclib, abemaciclib) and CDKN2A-loss-enriched trial arms.	test info · 500 Chipeta Way, Salt Lake City, UT 84108 · 1-800-242-2787
p16 IHC (clone E6H4 or equivalent)	NeoGenomics Laboratories	CDK4/6-inhibitor eligibility (palbociclib, ribociclib, abemaciclib) and CDKN2A-loss-enriched trial arms.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907
p16 IHC (clone E6H4 or equivalent)	LabCorp / Esoterix Oncology	CDK4/6-inhibitor eligibility (palbociclib, ribociclib, abemaciclib) and CDKN2A-loss-enriched trial arms.	test info · 358 South Main Street, Burlington, NC 27215 · 1-800-345-4363
p16 IHC (clone E6H4 or equivalent)	Quest Diagnostics	CDK4/6-inhibitor eligibility (palbociclib, ribociclib, abemaciclib) and CDKN2A-loss-enriched trial arms.	test info · 500 Plaza Drive, Secaucus, NJ 07094 · 1-866-697-8378

CCND3 alteration-class refinement (amplification vs activating mutation vs fusion) via NGS plus FISH or copy-number array as needed	Foundation Medicine (preferred) (FoundationOne CDx)	CDK4/6-inhibitor and cyclin-D-degrader trial enrichment (PF-07220060, fadraciclib, INCB123667).	test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639
CCND3 alteration-class refinement	Caris Life Sciences (Caris Molecular Intelligence)	CDK4/6-inhibitor and cyclin-D-degrader trial enrichment (PF-07220060, fadraciclib, INCB123667).	test info · 4610 South 44th Place, Phoenix, AZ 85040 · 1-888-979-8669
CCND3 alteration-class refinement	Tempus (Tempus xT)	CDK4/6-inhibitor and cyclin-D-degrader trial enrichment (PF-07220060, fadraciclib, INCB123667).	test info · 600 West Chicago Avenue, Chicago, IL 60654 · 1-800-739-4137
CCND3 alteration-class refinement	Memorial Sloan Kettering Diagnostic Molecular Pathology (MSK-IMPACT)	CDK4/6-inhibitor and cyclin-D-degrader trial enrichment (PF-07220060, fadraciclib, INCB123667).	test info · 1275 York Avenue, New York, NY 10065 · 1-212-639-2000
CCND3 alteration-class refinement	NeoGenomics Laboratories (NeoTYPE Comprehensive Panel)	CDK4/6-inhibitor and cyclin-D-degrader trial enrichment (PF-07220060, fadraciclib, INCB123667).	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907
Comprehensive tumor NGS including SMAD4, KEAP1, STK11, GNAS, ARID1A	Foundation Medicine (preferred) (FoundationOne CDx)	Frames prognosis under multi-agent therapy and weights any ICI-combo strategy on the KRAS axis.	test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639
Comprehensive tumor NGS	Caris Life Sciences (Caris Molecular Intelligence)	Frames prognosis under multi-agent therapy and weights any ICI-combo strategy on the KRAS axis.	test info · 4610 South 44th Place, Phoenix, AZ 85040 · 1-888-979-8669
Comprehensive tumor NGS	Tempus (Tempus xT)	Frames prognosis under multi-agent therapy and weights any ICI-combo strategy on the KRAS axis.	test info · 600 West Chicago Avenue, Chicago, IL 60654 · 1-800-739-4137
Comprehensive tumor NGS	NeoGenomics Laboratories (NeoTYPE Comprehensive Panel)	Frames prognosis under multi-agent therapy and weights any ICI-combo strategy on the KRAS axis.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907
Comprehensive tumor NGS	Memorial Sloan Kettering Diagnostic Molecular Pathology (MSK-IMPACT)	Frames prognosis under multi-agent therapy and weights any ICI-combo strategy on the KRAS axis.	test info · 1275 York Avenue, New York, NY 10065 · 1-212-639-2000
Germline hereditary-cancer panel covering BRCA1, BRCA2, PALB2, ATM, MLH1, MSH2, MSH6, PMS2, EPCAM, CDKN2A, TP53, STK11	Invitae (preferred) (Invitae Multi-Cancer Panel)	Olaparib maintenance after platinum for germline BRCA-mutant PDAC; family cascade testing; FAMMM-CDKN2A screening.	test info · 1400 16th Street, San Francisco, CA 94103 · 1-800-436-3037

Germline hereditary-cancer panel	GeneDx (GeneDx OncoGeneDx)	Olaparib maintenance after platinum for germline BRCA-mutant PDAC; family cascade testing; FAMMM-CDKN2A screening.	test info · 207 Perry Parkway, Gaithersburg, MD 20877 · 1-888-729-1206
Germline hereditary-cancer panel	Ambry Genetics (CancerNext)	Olaparib maintenance after platinum for germline BRCA-mutant PDAC; family cascade testing; FAMMM-CDKN2A screening.	test info · 1 Enterprise, Aliso Viejo, CA 92656 · 1-866-262-7943
Germline hereditary-cancer panel	Myriad Genetics (MyRisk Hereditary Cancer)	Olaparib maintenance after platinum for germline BRCA-mutant PDAC; family cascade testing; FAMMM-CDKN2A screening.	test info · 320 Wakara Way, Salt Lake City, UT 84108 · 1-800-469-7423
Germline hereditary-cancer panel	Color Health	Olaparib maintenance after platinum for germline BRCA-mutant PDAC; family cascade testing; FAMMM-CDKN2A screening.	test info · 831 Mitten Road, Burlingame, CA 94010 · 1-844-352-6567
MMR IHC panel (MLH1, MSH2, MSH6, PMS2)	Mayo Clinic Laboratories (preferred)	Confirms the MSS call; closes the door on tumor-agnostic pembrolizumab.	test info · 200 First Street SW, Rochester, MN 55905 · 1-800-533-1710
MMR IHC panel	ARUP Laboratories	Confirms the MSS call; closes the door on tumor-agnostic pembrolizumab.	test info · 500 Chipeta Way, Salt Lake City, UT 84108 · 1-800-242-2787
MMR IHC panel	LabCorp / Esoterix Oncology	Confirms the MSS call; closes the door on tumor-agnostic pembrolizumab.	test info · 358 South Main Street, Burlington, NC 27215 · 1-800-345-4363
MMR IHC panel	Quest Diagnostics	Confirms the MSS call; closes the door on tumor-agnostic pembrolizumab.	test info · 500 Plaza Drive, Secaucus, NJ 07094 · 1-866-697-8378
MMR IHC panel	NeoGenomics Laboratories	Confirms the MSS call; closes the door on tumor-agnostic pembrolizumab.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907
TMB assay platform and calling-pipeline confirmation (panel size, filters, harmonization with FDA-approved 10 mut/Mb threshold)	Foundation Medicine (preferred) (FoundationOne CDx (TMB-harmonized))	Confirms sub-threshold TMB and the foreclosure of tumor-agnostic pembrolizumab eligibility.	test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639
TMB assay platform confirmation	Caris Life Sciences (Caris Molecular Intelligence)	Confirms sub-threshold TMB and the foreclosure of tumor-agnostic pembrolizumab eligibility.	test info · 4610 South 44th Place, Phoenix, AZ 85040 · 1-888-979-8669
TMB assay platform confirmation	Tempus (Tempus xT)	Confirms sub-threshold TMB and the foreclosure of tumor-agnostic pembrolizumab eligibility.	test info · 600 West Chicago Avenue, Chicago, IL 60654 · 1-800-739-4137

TMB assay platform confirmation	NeoGenomics Laboratories (NeoTYPE Comprehensive Panel)	Confirms sub-threshold TMB and the foreclosure of tumor-agnostic pembrolizumab eligibility.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907
TMB assay platform confirmation	Memorial Sloan Kettering Diagnostic Molecular Pathology (MSK-IMPACT)	Confirms sub-threshold TMB and the foreclosure of tumor-agnostic pembrolizumab eligibility.	test info · 1275 York Avenue, New York, NY 10065 · 1-212-639-2000
TP53 variant annotation (LOF vs dominant-negative vs gain-of-function) with reference to IARC TP53 database	Memorial Sloan Kettering Diagnostic Molecular Pathology (preferred) (MSK-IMPACT (with variant annotation))	Frames any TP53-restoration or MDM2-inhibitor strategy; does not gate the KRAS axis.	test info · 1275 York Avenue, New York, NY 10065 · 1-212-639-2000
TP53 variant annotation	Foundation Medicine (FoundationOne CDx)	Frames any TP53-restoration or MDM2-inhibitor strategy; does not gate the KRAS axis.	test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639
TP53 variant annotation	Tempus (Tempus xT)	Frames any TP53-restoration or MDM2-inhibitor strategy; does not gate the KRAS axis.	test info · 600 West Chicago Avenue, Chicago, IL 60654 · 1-800-739-4137
TP53 variant annotation	Caris Life Sciences (Caris Molecular Intelligence)	Frames any TP53-restoration or MDM2-inhibitor strategy; does not gate the KRAS axis.	test info · 4610 South 44th Place, Phoenix, AZ 85040 · 1-888-979-8669
TP53 variant annotation	IARC TP53 Database (annotation reference)	Frames any TP53-restoration or MDM2-inhibitor strategy; does not gate the KRAS axis.	test info · 150 Cours Albert Thomas, 69372 Lyon CEDEX 08, France
p16 IHC across multiple tumor regions (primary site vs metastatic biopsy if available)	Mayo Clinic Laboratories (preferred)	Refines confidence in the CDKN2A loss call for CDK4/6-inhibitor strategy; does not gate enrollment.	test info · 200 First Street SW, Rochester, MN 55905 · 1-800-533-1710
p16 IHC multi-region	ARUP Laboratories	Refines confidence in the CDKN2A loss call for CDK4/6-inhibitor strategy; does not gate enrollment.	test info · 500 Chipeta Way, Salt Lake City, UT 84108 · 1-800-242-2787
p16 IHC multi-region	NeoGenomics Laboratories	Refines confidence in the CDKN2A loss call for CDK4/6-inhibitor strategy; does not gate enrollment.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907
p16 IHC multi-region	Memorial Sloan Kettering Diagnostic Molecular Pathology	Refines confidence in the CDKN2A loss call for CDK4/6-inhibitor strategy; does not gate enrollment.	test info · 1275 York Avenue, New York, NY 10065 · 1-212-639-2000
p16 IHC multi-region	LabCorp / Esoterix Oncology	Refines confidence in the CDKN2A loss call for CDK4/6-inhibitor strategy; does not gate enrollment.	test info · 358 South Main Street, Burlington, NC 27215 · 1-800-345-4363
Serial KRAS G12R ctDNA quantitation (ddPCR or tumor-informed MRD) at baseline and at cycle 2 / cycle 4 of KRAS-directed therapy	Natera (preferred) (Signatera (tumor-informed MRD))	Early-response and emerging-resistance surveillance on pan-RAS / RAS(ON) inhibitor therapy.	test info · 13011 McCallen Pass, Austin, TX 78753 · 1-650-249-9090

Serial KRAS G12R ctDNA quantitation	Guardant Health (Guardant Reveal / Guardant360 Response)	Early-response and emerging-resistance surveillance on pan-RAS / RAS(ON) inhibitor therapy.	test info · 505 Penobscot Drive, Redwood City, CA 94063 · 1-855-698-8887
Serial KRAS G12R ctDNA quantitation	Bio-Rad / academic ddPCR core	Early-response and emerging-resistance surveillance on pan-RAS / RAS(ON) inhibitor therapy.	test info · 1000 Alfred Nobel Drive, Hercules, CA 94547 · 1-510-724-7000
Serial KRAS G12R ctDNA quantitation	Tempus (Tempus xM)	Early-response and emerging-resistance surveillance on pan-RAS / RAS(ON) inhibitor therapy.	test info · 600 West Chicago Avenue, Chicago, IL 60654 · 1-800-739-4137
Serial KRAS G12R ctDNA quantitation	Foundation Medicine (FoundationOne Tracker)	Early-response and emerging-resistance surveillance on pan-RAS / RAS(ON) inhibitor therapy.	test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639
Tumor NGS for KRAS copy-number (KRAS amplification, wildtype KRAS amplification, secondary KRAS mutations)	Foundation Medicine (preferred) (FoundationOne CDx)	Frames durability and next-line planning for pan-RAS / RAS(ON) inhibitor therapy; does not gate enrollment.	test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639
Tumor NGS for KRAS copy-number	Caris Life Sciences (Caris Molecular Intelligence)	Frames durability and next-line planning for pan-RAS / RAS(ON) inhibitor therapy; does not gate enrollment.	test info · 4610 South 44th Place, Phoenix, AZ 85040 · 1-888-979-8669
Tumor NGS for KRAS copy-number	Tempus (Tempus xT)	Frames durability and next-line planning for pan-RAS / RAS(ON) inhibitor therapy; does not gate enrollment.	test info · 600 West Chicago Avenue, Chicago, IL 60654 · 1-800-739-4137
Tumor NGS for KRAS copy-number	Memorial Sloan Kettering Diagnostic Molecular Pathology (MSK-IMPACT)	Frames durability and next-line planning for pan-RAS / RAS(ON) inhibitor therapy; does not gate enrollment.	test info · 1275 York Avenue, New York, NY 10065 · 1-212-639-2000
Tumor NGS for KRAS copy-number	NeoGenomics Laboratories (NeoTYPE Comprehensive Panel)	Frames durability and next-line planning for pan-RAS / RAS(ON) inhibitor therapy; does not gate enrollment.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907
Tumor microenvironment workup: PD-L1 IHC (22C3), CD3 / CD8 TIL density, and stromal density (alpha-SMA or H and E review)	NeoGenomics Laboratories (preferred) (MultiOmyx multiplex IHC)	Frames ICI-combination strategy (KRAS-inhibitor plus PD-1, CXCR4 antagonist plus PD-1, vaccine plus PD-1); does not gate enrollment.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907
Tumor microenvironment workup	Caris Life Sciences (Caris Molecular Intelligence)	Frames ICI-combination strategy (KRAS-inhibitor plus PD-1, CXCR4 antagonist plus PD-1, vaccine plus PD-1); does not gate enrollment.	test info · 4610 South 44th Place, Phoenix, AZ 85040 · 1-888-979-8669

Tumor microenvironment workup	Foundation Medicine (FoundationOne CDx (with PD-L1 IHC reflex))	Frames ICI-combination strategy (KRAS-inhibitor plus PD-1, CXCR4 antagonist plus PD-1, vaccine plus PD-1); does not gate enrollment.	test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639
Tumor microenvironment workup	Mayo Clinic Laboratories	Frames ICI-combination strategy (KRAS-inhibitor plus PD-1, CXCR4 antagonist plus PD-1, vaccine plus PD-1); does not gate enrollment.	test info · 200 First Street SW, Rochester, MN 55905 · 1-800-533-1710
Tumor microenvironment workup	Memorial Sloan Kettering Diagnostic Molecular Pathology	Frames ICI-combination strategy (KRAS-inhibitor plus PD-1, CXCR4 antagonist plus PD-1, vaccine plus PD-1); does not gate enrollment.	test info · 1275 York Avenue, New York, NY 10065 · 1-212-639-2000

Decision support, not medical advice. Confirm assay availability and current standards with the treating team and the local pathology service.